MIND MATRIX

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SESSION: 08

REPORT ON: TASK REPORT

TASK NO: 02

TITLE:

ART SPACE APP DESIGN, STATES & TEST PLAN

1. Feature Description

The Art Space application allows users to browse artwork images and view details about each artwork. Users can also save artworks as favorites.

2. State Variables

State Variable	Type	Initial Value	Purpose
currentArtworkIndex	Int	0	Tracks current artwork
favouritesList	List	Empty	Stores favourite artworks
showDetails	Boolean	false	Controls details screen

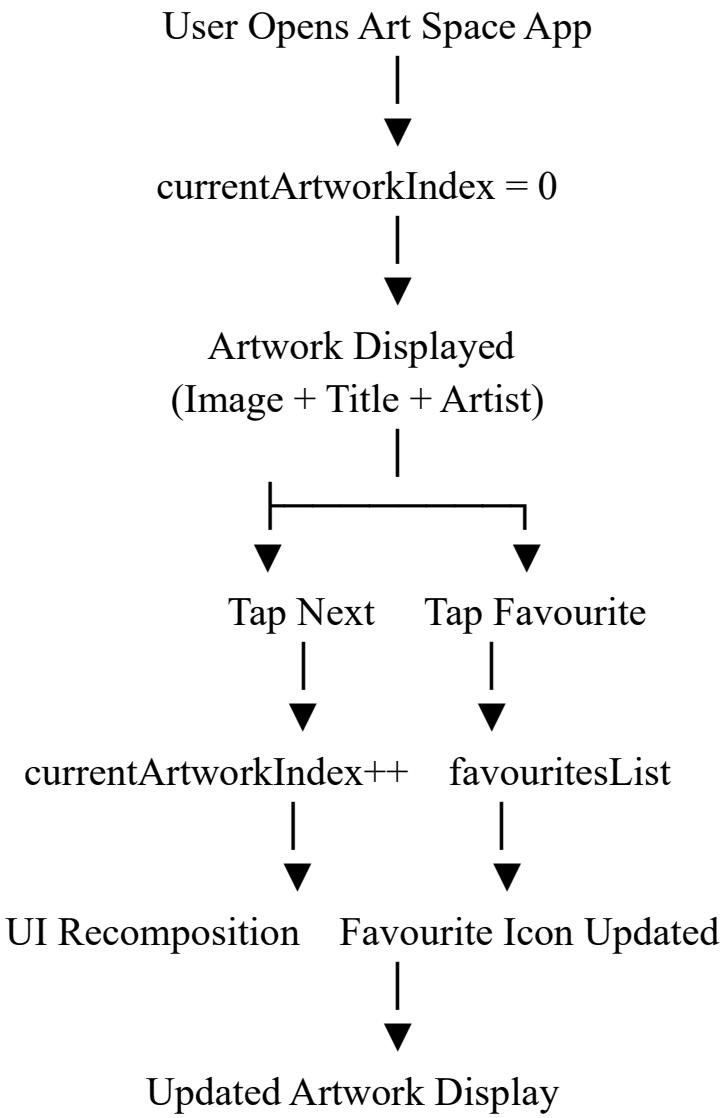
State Transitions:

- Tap **Next** → increase currentArtworkIndex

- Tap **Favourite** → add artwork to favouritesList

3. Application State Flow Diagram

Figure 2: Art Space UI State Transition



4. Testable Units

Unit	UI Component	Trigger	Expected Result
Next Artwork	Button	Click Next	Image changes
Favourite Toggle	Icon Button	Tap heart	Added to favourites
Details Screen	Text/Image	Tap artwork	Details appear

5. Test Plan Table

Test ID	Unit	Test Type	Preconditions	Trigger	Expected Outcome
T1	Next Button	UI Test	App running	Click Next	Next artwork shown
T2	Favourite Button	UI Test	Artwork visible	Tap favourite	Added to list
T3	Details View	UI Test	Artwork selected	Tap artwork	Details displayed
T4	Edge Case	UI Test	Last artwork	Press Next	Loop or disable
T5	Orientation Change	UI Test	App open	Rotate screen	State preserved

6. Reflection

Planning tests along with UI design improves reliability and helps identify bugs early. Testing UI interactions ensures that application state updates correctly. One challenge is handling UI synchronization and ensuring tests remain stable across different device states.